Kiwon Eom 엄기원

Undergraduate researcher, AIM Lab, Hanbat National University
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Research interests

Computer vision, multimodal learning, deep learning, machine learning, object detection, satellite image processing

Education

B.S. in Computer Engineering, Hanbat National University, Daejeon, Korea · Mar 2023 – present

Research experience

Undergraduate researcher, Artificial Intelligence Multimodal Lab (AIM Lab), Hanbat National University · Advisor: Prof. Haneol Jang · Feb 2024 – present

- AI and computer vision research, mainly on object detection and satellite image processing, focusing on tiny and densely packed objects in large-scale satellite and aerial imagery.
- Developing DETR- and YOLO-based detectors that preserve high-resolution features and suppress image noise, evaluated on public benchmarks through controlled comparisons.

Publications

† equal contribution

Kiwon Eom, Eunkyung Kim, Haneol Jang. **YOLO-Dos: Improving YOLO-based SAR object detection performance via a speckle suppression module.** Journal of IKEEE, 30(1):171-181, 2026 · KCI-indexed (in Korean). DOI

Kiwon Eom, Dong-Su Kim, Tae-Woo Oh, Haneol Jang. **ViT-based deepfake detection via regional emphasis weights on eyes, nose, and mouth.** Journal of Digital Forensics, 19(4):116-131, 2025 · KCI-indexed (in Korean). DOI

Kiwon Eom†, Cheoneum Park†, Eunkyung Kim, Haneol Jang. **Tiny Object Detection in Aerial Images.** Scientific Reports (SCIE, IF 4.9, Top JCR 14.6%), 2026 · under review.

Gyeom Kim, **Kiwon Eom**, Haneol Jang. **Improving Tiny Object Detection via High-Resolution Feature Refinement.** Journal of IKEEE, 2026 · under review.

Gyeom Kim, **Kiwon Eom**, Seongmin Pyo, Haneol Jang. **Density-Aware Query Score Adjustment for Tiny Object Detection.** KIBME2026, Jun 2026.

Gyeom Kim, **Kiwon Eom**, Haneol Jang. **Improving Tiny Object Detection through High-Frequency Enhancement of High-Resolution Features.** KIBME2026, Nov 2026 · under review.

Kiwon Eom, Haneol Jang. **FUSE-DETR: DETR with Frequency-guided Unified Scale Enhancement for Tiny Object Detection.** KNST 2026 Summer Conference, Jul 2026.

Kiwon Eom, Haneol Jang. **Tiny Object Detection in Satellite Images via 4-Level Feature Pyramid Extension of DETR-Based Models.** IEIE Summer Conference 2026, Jun 2026.

Kiwon Eom, Eunkyung Kim, Haneol Jang. **YOLO-Dos: Enhancing YOLO-based SAR Object Detection via a Speckle Suppression Module.** KIBME 2025 Winter Conference, Dec 2025. DBpia

Kiwon Eom, Dong-Su Kim, Haneol Jang. **Enhancing ViT-Based Deepfake Detection by Emphasizing Eye, Nose, and Mouth Patches.** KIBME 2025 Summer Conference, Jun 2025. DBpia

Honors and competitions

Top 3% (29th/1,090), LG Aimers 9th cohort hackathon, online round · finalist (96 of 3,393), 18th/33 in the final · 2026

Top 5% (6th/138), National Park Satellite Monitoring AI Challenge · coastal debris · 2026

Top 6% (16th/271), SW-Centered University Digital Competition · AI-generated text detection · 2025

Top 9% (85th/985), Wind Power Forecasting AI Competition (BARAM 2026) · 2026

Top 13% (28th/219), SW-Centered University Digital Competition · fake voice detection · 2024

1st place, cloud segmentation Kaggle challenge (HBNU AI course) · 2025

1st place, image classification (face anti-spoofing) Kaggle challenge (HBNU AI course) · 2025

Projects

Intelligent processing of puzzle-type ultra-wide-area satellite imagery · KRIT Specialized Research Lab, Sub-project 5 · dataset analysis, DETR-based network design, evaluation · 2026 – present

AI-based detection of unpermitted livestock barns (with K-water) · barn detection models for a K-water pilot service · 2026 – present

Tiny object detection in satellite and aerial imagery (capstone design, TOD team) · web demo development and deployment · 2026 – present

Skills

Tools Python, C, Java · PyTorch, MMDetection, Ultralytics YOLO, OpenCV · Linux, Docker, Git, W&B